Introduction to Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) is a technique that improves the answers of a large language model by giving it relevant information retrieved from an external knowledge source at query time. Instead of relying only on what the model memorised during training, a RAG system first searches a document collection and then asks the model to answer using the retrieved passages.

How a RAG pipeline works

A typical pipeline has four steps. First, documents are split into small overlapping chunks. Second, each chunk is converted into an embedding, a numeric vector that captures its meaning. Third, the embeddings are stored in a vector database such as ChromaDB or FAISS. Fourth, when a user asks a question, the question is embedded, the most similar chunks are retrieved using similarity search, and they are passed to the language model as context.

Why use RAG

RAG reduces hallucination because answers are grounded in real documents. It lets a company keep its knowledge up to date without retraining the model, and it allows answers to cite the source page. The main trade-offs are extra latency from retrieval and the need to tune chunk size and the number of retrieved chunks.